

2025-11-04-strathclyde OpenRefine lesson instructor notes

Summary and Setup

[SLIDE HERE: Links/QR codes]

- Lesson site: <https://swcarpentry.github.io/shell-novice/>

Summary

- Before you can analyse your data, you normally need to *clean it**
- What is "data cleaning"?
 - Identification (and maybe correction) of errors in the data
 - Formatting data for consistency
- **If your data is not clean, your analysis results may be false and/or not reproducible**
- OpenRefine is a powerful, free, and open source tool for working with messy data
 - It can clean data and transform it from one format to another
 - It can also generate reproducible data cleaning workflows or pipelines that track the changes you make
 - The scripts can also be applied to **undo** changes that were made to the raw data
- Data cleaning steps often need to be applied to more than one file
 - The pipelines OpenRefine makes allow you to reapply the processes to new datasets

Introduction

- OpenRefine is a Java program that runs on your own machine
 - It is a desktop application that uses your web browser as an interface
 - No internet connection is required and no data or commands are sent to any server: **it can be used on confidential data**
- OpenRefine **does not modify your original dataset** - this is best practice
 - You must save the cleaned data in a different file
- All OpenRefine actions can be reversed
- OpenRefine saves as you go - you can leave or pick up a data cleaning project at any point
- OpenRefine can import your data in many formats, including
 - Comma-separated (CSV) or tab-separated (TSV)
 - text files

- fixed-width columns
- JSON
- XML
- OpenDocument Spreadsheet (ODS)
- Excel spreadsheet (XLS/XLSX)
- RDF data

Importing data

[OPENREFINE DEMO]

- Let's get started with OpenRefine, by creating a project
- Start OpenRefine
 - This will open in a web browser at <http://127.0.0.1:3333>
 - In the left margin you'll see options: **Create Project**, **Open Project** and so on
- Click **Create Project**
 - The window "Create a project by importing data" should be visible
- Select **This Computer** as you want to use a local file
- Click **Browse**
 - Navigate to where you downloaded **Portal_rodents_19772002_simplified.csv**
 - Select that file and click **Open**
 - The filename should be visible
- Click **Next** to upload the data to OpenRefine
 - Once it's loaded, OpenRefine will present a preview of your data
- **Any obvious errors (like file formatting problems) should be visible at this point**
- In the middle of the page there should be a set of options including **Character Encoding**
 - Ensure that the **Trim leading and trailing whitespace from strings** option is **not** checked
- Everything should look OK
- Click **Create project**
- OpenRefine should present you with a table of 790 rows
 - All data is imported as text - **even numerical data**

Exploring data

Exploring data with facets

- **Facets** are a useful concept in OpenRefine

- **Faceting** is the process of exploring data using filters, to explore its composition
- Faceting allows us to identify a subset of data to change in bulk
- **Faceting groups all like values that appear in a column**
 - It allows filtering by those values
- This is easier to understand through practice
- Find the **scientificName** column in the dataset
- Click the down arrow/triangle next to the header
- Choose **Facet -> Text facet**
- In the left panel you will see a box representing each unique value in the **scientificName** column
 - The number of times each entry appears is also shown
- Sort the facet by **name** and **count**
 - What problems do you see with the data?
- Hover your mouse over a name in the facet
 - You should see an **edit** and an **include** link
 - If you click on **edit** you will see options to repair errors immediately - **don't do this yet**

[SHOW SLIDES: Challenge (faceting)]

[OPENREFINE DEMO]

- By default, all imported columns are treated as text
- We can convert data we know to be a different data type (e.g. number or date) using **Edit cells -> Common transforms**
 - **Sometimes the data has errors that prevent us treating it as numeric** so we explore it with a facet first
- Click on the down arrow/triangle for **yr**
- Select **Facet -> Numeric facet**
 - This will generate a facet that says there is no numeric data present (the data is in text format)
- Click on the down arrow/triangle for **yr**
- Select **Edit cells -> Common transforms... -> To number**
 - The data shifts to right-justified and turns green
 - The facet shows a histogram
- Transform **period**, **plot**, and **recordID** from text to numbers

- Remove the facets by clicking the X

Using scatterplots

- We can see how numeric columns relate to each other using the **scatterplot facet**
- Click on the down arrow/triangle for `recordID`
 - Select **Facet -> Scatterplot facet**
 - A new window called **Scatterplot matrix** appears
 - This plots each numeric column against the others
- Select the plot of `recordID` against `yr`
 - A facet of the comparison appears
- Click in the scatterplot and drag to highlight a rectangle
 - **What happens to the data?** It is selected/subsetted in the table
 - **We can use this approach to identify outliers in a dataset**
- Click anywhere in the scatterplot to deselect

Transforming data

Data splitting

- We can split data from one column into multiple columns, if we can identify a common separator (like a comma or space)
- Let's split the `scientificName` into two columns: one for genus, one for species
- Click on the down arrow/triangle for `scientificName`
 - Select **Edit Column -> Split into several columns...**
 - In the **Separator box** replace the comma with a space
 - **Uncheck the Remove this column box**
 - Click **OK**
- You should now see several columns: `scientificName1`, `scientificName2`, etc.
 - We can see that some of the data was not well-formatted, in the sense it wasn't simple *Genus species* data
 - We're going to have to fix this

Undoing/redoin actions

- But first let's undo what we just did with the split
 - This is quite a common requirement when cleaning data
- Click **Undo/Redo** in the left margin
 - You should see all the changes you made to the data in that panel
 - The current stage is highlighted
 - We can click on any stage in the processing to jump to that version of the data
- Click on the appropriate step just before splitting `scientificName`

- We could click on the last stage and redo the change, but let's not do that.

Trimming leading/trailing whitespace

- It's quite common for whitespace to be added at the start/end of text in datasets
 - But it's difficult to spot this by eye
- OpenRefine can remove these characters for us automatically
- Click on the down arrow/triangle for `scientificName`
 - Select `Edit cells -> Common transforms -> Trim leading and trailing whitespace`
 - OpenRefine will warn us about deleting project history (taking a different path), but click `Apply anyway`
- Let's split the `scientificName` into columns again: one for genus, one for species
- Click on the down arrow/triangle for `scientificName`
 - Select `Edit Column -> Split into several columns...`
 - In the `Separator box` replace the comma with a space
 - **Uncheck the Remove this column box**
 - Click `OK`
- You should now see two columns: `scientificName1`, `scientificName2`

Renaming columns

- We have Genus and species separated into two columns, but they have generic names
 - We want to call the columns `genus` and `species`
- Click on the down arrow/triangle for `scientificName1`
 - Select `Edit Column -> Rename column...`
 - Enter `genus` into the field
- Click on the down arrow/triangle for `scientificName2`
 - Select `Edit Column -> Rename column...`
 - Enter `species` into the field
 - **You will not be allowed to rename the column**
 - **Change the name of the `species` column to `species_abbreviation`**
 - Rename the `scientificName2` column to `species`

Combine columns

- The date for each row is split in three columns, for month, day, and year
 - **We want to create a single column which contains the date**

- Click on the menu for the **yr** column
 - Select **Edit column -> Join columns...**
 - Check the boxes next to **mo** and **dy**
 - Enter **-** as a separator
 - Select the option **Write result in new column named** and write **date** in the field
 - Click **OK**
- Click on the menu for the new **date** column
 - Select **Edit cells -> Common transforms -> To date**
- Now we can use a **Timeline facet** to inspect the data
- Click on the menu for the new **date** column
 - Select **Facet -> Timeline facet**
 - A histogram view of sampled dates should appear in the left margin

Data clustering

- Clustering allows us to find groups of entries that are not identical, but that are similar
- These may be alternative representations of the same thing - possibly even typos
- Make a text facet for **scientificName**
- In **scientificName** there are several near-identical entries
 - For example a misspelling of *Ammospermophilus harrisii*
 - Click the **Cluster** button
 - Try some options for **Method** and **Keying function** and click the **Cluster** button
- Choose **Key collision** and **Metaphone3**
 - This identifies a cluster of options for *Ammospermophilus harrisii*
 - It also suggests a new cell value to replace the likely misspellings
- Tick the **Merge** checkbox
- Click **Merge selected and close**
- Note that the facet has updated so that there are no more misspellings of *Ammospermophilus harrisii*

Filtering and sorting

- Sometimes you only want to view and work with a subset of your data, or to apply an operation only to that subset
- In OpenRefine, we do this with **filters**

Include/Exclude in facets

- The **include** and **exclude** buttons in facets allow us to control our subset for working with

- In the text facet for **scientificName**, hover over *Baiomys taylori*
 - The **edit** and **include** options will appear
 - Click on **include**
- The *Baiomys taylori* text changes to orange to show it is included, and the **exclude** option appears
 - The number of matching rows has also changed (to 33 from 790)
- Hover over *Chaetodipus baileyi* and click **include**
 - This text also turns orange, and the number of matching rows increases to 35
- Click on **reset** at the top of the facet
 - This returns the selection to the original state
 - All the entries in the facet return to blue text

Text filters

- We can also create text filters on columns
- Close all the facets that are currently open
- Click on the down arrow/triangle for **scientificName**
 - Click on **Text filter**
 - A new text filter opens in the left margin
- Type **bai** into the filter field
 - The top of the page shows the 35 rows where the **scientificName** column contains the string **bai**
- Change the number of rows to be shown to **50**
 - Scroll down to see the matching rows
- Click on the down arrow/triangle for **scientificName**
 - Click on **Facet -> Text facet**
 - We can see that two species have been selected

Sorting data

- Sorting data is a good way to see the range spanned by your data, and to spot outliers at the extremes
- The **Sort...** tool is available in the drop-down menu for each column
- Click on the down arrow/triangle for **mo** (month)
 - Click on **Sort...**
 - Select **numbers** and make sure **smallest first** is checked so that months are sorted in order
 - Click **OK**
- Click on the down arrow/triangle for **plot**

- Click on **Sort...**
- Select **numbers** and make sure **smallest first** is checked
- Check **sort by this column alone** to see which samples came from plot 1 and clear the sorting on **mo**
- We can also sort by multiple columns
- Sorting takes place in the order you apply **sort** to each column
- Click on the down arrow/triangle for **yr** (year)
 - Click on **Sort...**
 - Select **numbers** and make sure **smallest first** is checked so that years are sorted in order
 - - Check **sort by this column alone** to see which samples came from plot 1 and clear the sorting on **plot**
 - Click **OK**
- Click on the down arrow/triangle for **mo** (month)
 - Click on **Sort...**
 - Select **numbers** and make sure **smallest first** is checked so that months are sorted in order
 - Click **OK**
- Click on the down arrow/triangle for **day** (day)
 - Click on **Sort...**
 - Select **numbers** and make sure **smallest first** is checked so that days are sorted in order
 - Click **OK**
- The data should now be in chronological order
- Click on **reset** in the **Facet / Filter** panel

Exporting, importing, and applying the data cleaning pipeline/workflow

- OpenRefine saves every change you make during your data cleaning and preliminary analysis
 - The data is kept in a file format called JSON - an open standard - that is shareable with other applications and which can be applied to other data files
 - If you find yourself having to clean multiple data files of the same kind, you can use this JSON script to clean them quickly in OpenRefine
- Click on the **Undo / Redo** tab
 - Click on the **Extract** button
 - This shows the entire history of operations you have performed on your data, in order
 - On the left you can choose to include some, none, or all of these steps in the exported file
 - The operations that will be exported are shown on the right hand side of the window (this is JSON format)
- Click on the **Export** button

- Save the file as **history.json** in a convenient location
- Click the **Close** button
- We can use this JSON file to clean a new dataset
 - Let's practice in a new project
- Click on the **Open** button at the top of the window
 - This opens a new project in the browser
- Click **Browse**
 - Navigate to where you downloaded **Portal_rodents_19772002_simplified.csv**
 - Select that file and click **Open**
 - The filename should be visible
- Click **Next** to upload the data to OpenRefine
 - Once it's loaded, OpenRefine will present a preview of your data
- In the middle of the page there should be a set of options including **Character Encoding**
 - Ensure that the **Trim leading and trailing whitespace from strings** option is **not** checked
- Everything should look OK
- Click **Create project**
 - You should recognise the uncleaned data from earlier
- Click the **Undo / Redo** tab
 - Click the **Apply** button
 - Click **Browse** and select the **history.json** file you just saved - the JSON script will appear in the window
 - Click the **Run Operations** button
- You should see a newly-cleaned dataset
- Using standardised data cleaning approaches is an essential component of reproducible science
 - These kinds of data-cleaning scripts should be provided alongside your published work and data (e.g. as Supplementary Information)

Exporting and saving cleaned data

- When you have cleaned your data, you will want to export it to a new file
- **It is important to keep raw and cleaned data in separate files**
- Click the **Export** button at the top right of the window

- Click on **Tab-separated value**
 - In the file dialogue box, change the filename to something that indicates the data have been cleaned (e.g. **Portal-rodents-19772002-simplified-csv-cleaned.tsv**)
 - Click on **Save**
- The file can then be imported into downstream analysis tools

Saving/Exporting a project

- In addition to saving the data cleaning pipeline and the cleaned data separately, OpenRefine can save the entire project
- OpenRefine saves your projects by default
 - When you start OpenRefine you will be presented with a list of former projects
 - You can click on items in this list to open them
- You can also export the project, so it can be shared with others
- Click the **Export** button
 - Click **OpenRefine project archive to file**
 - In the file dialogue box, navigate to a convenient location, and click **Save**
- This saves a **.tar.gz** (compressed) file that can be reused by OpenRefine

Importing a project

- To import an OpenRefine project, click on the **Open** button
 - Click on **Import project** in the left menu
 - Click on **Browse** and navigate to the location of the project file
 - Select it and click **Open**
 - Click **Import project**
- You will be presented with a view onto the project, with the history of actions intact

Summary

- That's all I have to say about OpenRefine
 - You can find more resources listed at <https://datacarpentry.github.io/OpenRefine-ecology-lesson/10-resources.html>
- In this session you've learned how to use OpenRefine to reproducibly clean and inspect your data
 - You've created, imported, and exported projects
 - You've imported raw data and exported cleaned data
 - You've created, imported, and exported custom data-cleaning workflows
 - You've used facets to edit and clean data
 - You've used facets and filters to subset your data for analysis
 - You've used OpenRefine to transform your data into more useful forms (like combining or splitting columns, and clustering)

- This is a very useful skillset to have for professional, reproducible science and data analysis